# ME 5309 – Mid-Semester Examination: ANSWER SHEET

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**Questions attempted:** Q3, Q4

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## Question 3

*Given: A = [0, 1; 0, −2], B = [0; 1], C = [1, 0], $f_s$ = 1 kHz so the sample time is T = 1/$f_s$ = 1 ms.*

**(a)** Reason 1 – noise amplification: the backward difference $\hat{\omega}_k = (\theta_k - \theta_{k-1})/T$ has frequency response $(1 - e^{-j\omega T})/T$, whose magnitude grows with frequency (≈ ω for small ωT), i.e. it is a high-pass operation. Any noise on the encoder signal therefore appears in the velocity multiplied by roughly 1/T = 1000 $s^{-1}$, and can swamp the true velocity.

Reason 2 – quantisaton: the encoder only reports integer counts, so Δθ is an integer multiple of the count size q. Dividing by T = 1 ms gives velocity steps of q/T; for a 4096-count encoder q = $2\pi/4096 = 1.53\times10^{-3}$ rad, so one step is 1.53 rad/s. At low speed the estimate flips between 0 and one step (a noisy staircase), exactly where the damping term of the PD law is needed.

(A further, smaller issue is the half-sample delay of a backward difference, but the two reasons above are the main ones.)

**(b)** The analogue prototype is the first-order low-pass $H(s) = \omega_c/(s + \omega_c)$ with $\omega_c = 2\pi f_c = 2\pi \times 20$ = 125.66 rad/s. The bilinear (Tustin) transform replaces s by $(2/T)(z - 1)/(z + 1)$ with $T = 1/f_s$ = 0.001 s.

Define $a = \omega_c T/2 = 125.66 \times 0.001/2 = 0.062832$. Then $H(z) = \omega_c / [(2/T)(z-1)/(z+1) + \omega_c] = a(z + 1) / [(1 + a) z - (1 - a)] = a(1 + z^{-1}) / [(1 + a) - (1 - a) z^{-1}]$.

Dividing through by (1 + a): a/(1 + a) = 0.062832/1.062832 = 0.059117 and (1 − a)/(1 + a) = 0.937168/1.062832 = 0.881767. Hence with u = differenced signal and y = filtered velocity: $\mathbf{y_k = 0.05912\,(u_k + u_{k-1}) + 0.88177\,y_{k-1}}$. (Frequncy pre-warping is neglected since $f_c \ll f_s/2$.)

Verification of the DC gain: set u and y constant (z = 1): y = 2(0.05912)u + 0.88177 y ⇒ y(1 − 0.88177) = 0.11824 u ⇒ y/u = 0.11824/0.11823 = 1.000. In general H(1) = 2a/[(1 + a) − (1 − a)] = 2a/2a = 1 exactly, so the steady-state output equals the commanded input ✓.

**(c)** $A - LC = [-\ell_1, 1; -\ell_2, -2]$, so $\det(sI - A + LC) = (s + \ell_1)(s + 2) + \ell_2 = s^2 + (2 + \ell_1)s + (2\ell_1 + \ell_2)$.

Desired: $(s + 10)(s + 10) = s^2 + 10s + 100$. Matchng: $2 + \ell_1 = 10 \Rightarrow \ell_1 = 8$; $2(8) + \ell_2 = 100 \Rightarrow \ell_2 = 84$.

**$L = [8, 84]^T$**.

**(d)** Advantage of the observer that no filter of this kind can give: the observer is driven by both the measurement y and the model with the actual input u, so it predicts the motion between samples. The velocity estimate therefore has little or no phase lag. The low-pass filter on differenced data, by contrast, always pays a lag of about $1/(2\pi f_c)$ = 8 ms for its noise rejection, a trade-off it cannot escape. In addition the observer gives all states, and can be augmented with a disturbance state (e.g. unknown load torque) which no filter on θ can estimate.

Circumstance where the filter is safer: when the plant model is poorly known or changes during operation, for example an unknown payload, varying friction, contact with the environment, or actuator saturation making u differ from the commanded value. The observer propagates the wrong model so its estimate can be biased (and in the worst case unstable with the feedback loop), whereas the model-free filter just smooths what the encoder actually measures.

*Sampling at 1 kHz is fast compared with the 10 rad/s observer bandwidth, so discretisaton effects are small.*

## Question 4

*Given: plant $\theta\ddot{} + a\theta\dot{} + c\theta = b\,u$ with $a = 2$, $c = 6$, $b = 1$; controller $u = N_{ff}\,r - k_1\theta - k_2\theta\dot{}$.*

**(a)** Substitute the control law into the plant: $\theta\ddot{} + a\theta\dot{} + c\theta = b\,N_{ff}\,r - b\,k_1\,\theta - b\,k_2\,\theta\dot{}$. Moving the feedback terms to the left: $\theta\ddot{} + (a + b\,k_2)\,\theta\dot{} + (c + b\,k_1)\,\theta = b\,N_{ff}\,r$.

Numerically, $a + b\,k_2 = 2 + 2 = 4$ and $c + b\,k_1 = 6 + 2 = 8$, so the characteristic equation is $s^2 + 4s + 8 = 0$. Roots: $s = [-4 \pm \sqrt{(16 - 32)}]/2 = -2 \pm 2j$. They have negative real parts, so the loop is stable; $\omega_n = \sqrt{8} = 2.83$ rad/s and $\zeta = 4/(2 \times 2.83) = 0.707$.

Transfer function from r to θ: $T(s) = b\,N_{ff} / (s^2 + 4s + 8)$. DC gain ($s = 0$): $T(0) = N_{ff}/8$. For unity DC gain, $\mathbf{N_{ff} = 8}$. (Note that this is larger than 1 because the gravity-related stiffness c and the feedback $k_1$ both resist the displacement.)

**(b)** With the actual $b = 0.75$ and unchanged $N_{ff} = 8$, $k_1 = k_2 = 2$, the closed loop becomes $\theta\ddot{} + (a + bk_2)\theta\dot{} + (c + bk_1)\theta = bN_{ff}\,r$, i.e. $\theta\ddot{} + 3.5\theta\dot{} + 7.5\theta = 6\,r$.

For a unit step the steady state is found by setting derivatives to zero: $7.5\,\theta_{ss} = 6 \times 1$, so $\theta_{ss} = 6/7.5 = \mathbf{0.8}$.

Tracking error $= (r - \theta_{ss})/r \times 100\% = (1 - 0.8) \times 100 = \mathbf{20\%}$. A common mistake would be to say the output falls by the same 25% as the actuator gain; that is wrong here, becuase b also appears in the feedback stiffness $c + bk_1$, which decreases from 8 to 7.5 and partially compensates. The poles also change: $s^2 + 3.5s + 7.5 \rightarrow s = -1.75 \pm 2.11j$, still stabel.

**(c)** Take the state $z = [\theta, \theta\dot{}, \xi]^T$. The plant gives $\theta\ddot{} = -a\theta\dot{} - c\theta + b\,u$ with $u = -k_1\theta - k_2\theta\dot{} + K_I\xi$, and $\xi\dot{} = r - \theta$. With $r = 0$ (r only shifts the equilibrium) the closed-loop matrix is $A_{cl} = [0, 1, 0; -(c + bk_1), -(a + bk_2), bK_I; -1, 0, 0]$.

Characteristic polynomial $\det(sI - A_{cl})$, expanding along the first row: $s\cdot[(s + a + bk_2)s - 0] + 1\cdot[(c + bk_1)s + bK_I] = s^3 + (a + bk_2)s^2 + (c + bk_1)s + bK_I$. **(Result)**

Desiered poles $-4$ and $-2 \pm 2j$: $(s + 2 - 2j)(s + 2 + 2j) = (s + 2)^2 + 4 = s^2 + 4s + 8$. Multiply by $(s + 4)$: $s^3 + 4s^2 + 8s + 4s^2 + 16s + 32 = s^3 + 8s^2 + 24s + 32$.

Compare coefficients with $a = 2$, $b = 1$, $c = 6$: $s^2$: $2 + k_2 = 8 \Rightarrow \mathbf{k_2 = 6}$. $s^1$: $6 + k_1 = 24 \Rightarrow \mathbf{k_1 = 18}$. $s^0$: $b\,K_I = 32 \Rightarrow \mathbf{K_I = 32}$.

Check: all coefficients of the desired polynomial are positive and $(8)(24) = 192 > 32$, so Routh–Hurwitz confirms the three poles are in the left half plane ✓.

**(d)** Suppose the closed loop is stabel and r is constant. Then the state converges to a constant, so $\theta\dot{} = \theta\ddot{} = 0$ and, crucially, $\xi\dot{} = 0$ as well (ξ itself is constant at its final value $\xi_{ss}$). From the integrator equation $\xi\dot{} = r - \theta$, we get $r - \theta_{ss} = 0$, i.e. $\mathbf{\theta_{ss} = r}$.

Neither b nor c appears in this argument. They only influence the value of $\xi_{ss}$ needed to hold the equilibrium (from the plant equation $c\theta_{ss} = bu_{ss} = b(-k_1\theta_{ss} + K_I\xi_{ss})$), and the pole locations, so the zero error survives any error in the model parameters provided the loop stays stabel (unlike the feed-forward $N_{ff}$ scheme of part (b)).

Practical failure mode: integrator windup. If the actuator saturates (torque limit), the loop is effectively open, the error stays non-zero and ξ keeps integrating to a huge value. When the reference becomes reachable the stored integral forces a large overshoot and a long recovery. Standard remedy: anti-windup, such as clamping the integrator when the acuator is saturated, conditional integration, or back-calculation that feeds the difference between commanded and saturated torque back to the integrater.

*Note: the integrator adds one state, so the loop becomes third order.*